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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 11: Introduction to Python



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Learning Objectives

By the end of this lesson, students will be able to:

- Understand what Python is.
- Learn why Python is one of the world's most popular programming languages.
- Identify real-world uses of Python.
- Install and open Python (or an online Python editor).
- Write and run their first Python program.
- Understand basic Python syntax and comments.

Introduction

Have you ever wondered how:

- 🎮 Games are created?
- 🤖 Robots work?

- 💬 Chatbots answer questions?
- 🎥 YouTube recommends videos?
- 🚗 Self-driving cars make decisions?

The answer is **Programming**.

One of the easiest programming languages to learn is **Python**.

Python is simple, powerful, and fun, making it perfect for beginners.

What is Python?

Python is a **high-level programming language** used to create computer programs, websites, games, mobile apps, Artificial Intelligence (AI), Machine Learning (ML), and much more.

Simple Definition

Python is a language that helps us **give instructions to a computer**.

Just as we use English to talk to people, we use Python to "talk" to computers.

What is Programming?

Programming means writing **instructions (called code)** that tell a computer what to do.

Example

When you ask a computer to display:

Hello!

you write a Python program.

History of Python

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- Python was created by Guido van Rossum.
- Development began in **1989**.

- The first public version was released in **1991**.
- Python was named after the comedy show **"Monty Python's Flying Circus"**, not the snake.

Why Learn Python?

Python is:

- ✅ Easy to learn
- ✅ Easy to read
- ✅ Beginner-friendly
- ✅ Powerful
- ✅ Used around the world
- ✅ Free and open source
- ✅ Great for AI and Machine Learning

Features of Python

Feature	Meaning
Simple Syntax	Easy to read and write
Free	Anyone can download and use it
Cross-Platform	Works on Windows, macOS, and Linux
Open Source	Anyone can improve it
Powerful	Used for many types of projects
Large Community	Millions of programmers use Python

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Where is Python Used?





Python is used in many fields:

Artificial Intelligence (AI)

- Smart assistants
- Chatbots
- Image recognition

Data Science

- Data analysis
- Charts
- Reports

Web Development

- Websites
- Online applications

Game Development

- Educational games
- Puzzle games

Robotics

- Robot control
- Automation

Cloud Computing

- Online services
- Web applications

Installing Python

Students can use:

Option 1: Python Installed on the Computer

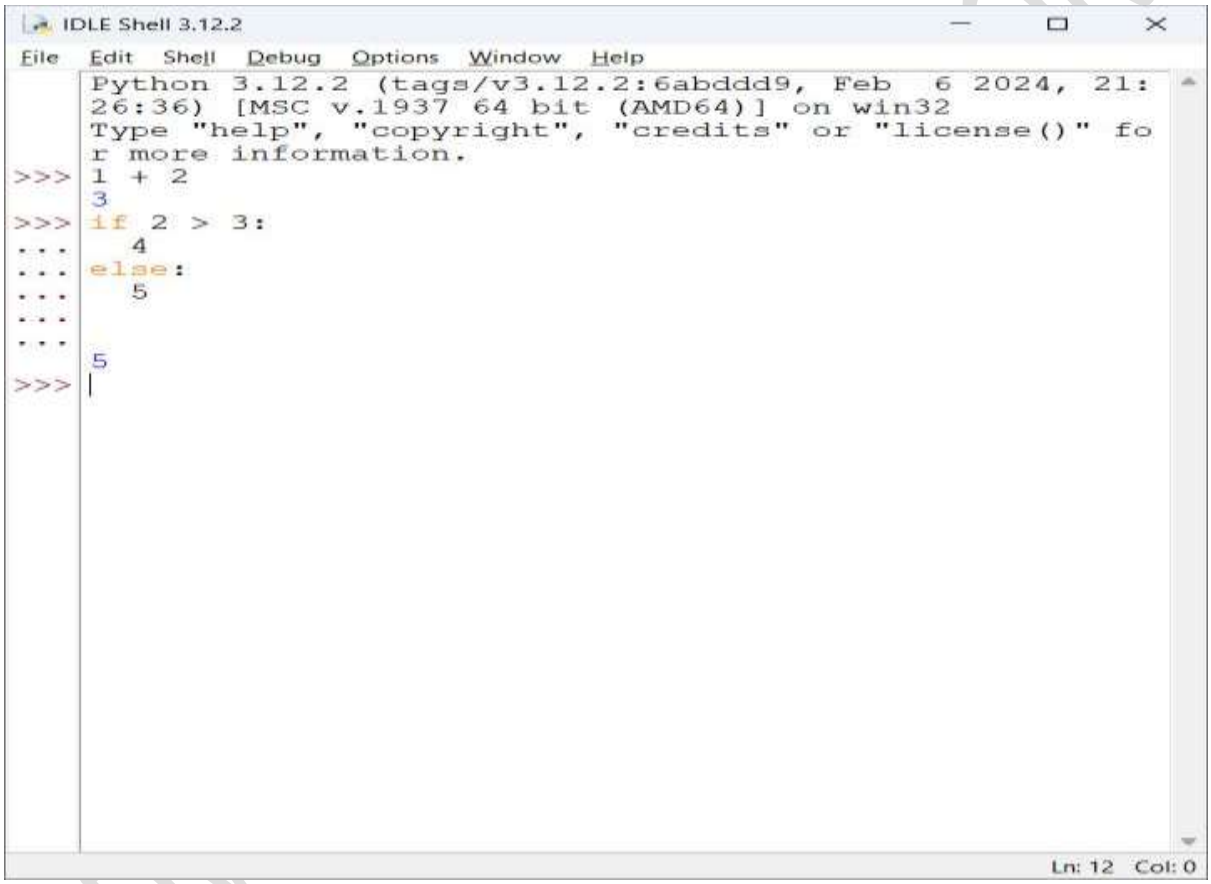
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Install Python and open the **IDLE** editor.

Option 2: Online Python Editor

Use a browser-based Python editor provided by your teacher or school.

Python IDLE







IDLE is the simple editor that comes with Python.

It helps us:

- Write code
- Run programs
- Test ideas
- Learn programming

Your First Python Program

Type the following code:

```
print("Hello, World!")
```

Output

```
Hello, World!
```

Explanation

- print() displays text on the screen.
- "Hello, World!" is the message to display.

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Printing Your Name

```
print("My Name is Rahul")
```

Output

```
My Name is Rahul
```

Replace **Rahul** with your own name.

Printing Multiple Lines

```
print("Welcome to Python")
print("I am learning programming.")
print("Python is fun!")
```

Output

```
Welcome to Python
I am learning programming.
Python is fun!
```

Comments in Python

Comments help explain the code.

Python ignores comments while running the program.

Example:

```
# This is my first Python program

print("Hello Students")
```

Python Rules

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- Python is **case-sensitive**.
- print is different from Print.
- Use brackets correctly.
- Check spelling carefully.
- Use quotation marks for text.

Common Mistakes

Incorrect	**Correct**
Print("Hello")	print("Hello")
print(Hello)	print("Hello")
Missing)	Close all brackets

Real-Life Examples of Python

Application	**Python Use**
School Projects	Simple programs
Games	Create games
AI	Chatbots and smart assistants
Science	Experiments and calculations
Robots	Robot control
Websites	Build web applications

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Practical Activities

Practical Activity 1 – Run Your First Program

Objective

Write and run your first Python program.

Steps

1. Open Python IDLE or the online editor.
2. Type:

```
print("Hello, World!")
```

3. Run the program.
4. Observe the output.

Practical Activity 2 – Print Personal Information

Type:

```
print("My Name is _____")
print("I study in Class 7/8")
print("My Favourite Subject is _____")
print("I want to become a _____")
```

Replace the blanks with your own information.

Practical Activity 3 – Print Your School Name

```
print("My School Name")
print("My City")
print("India")
```

Practical Activity 4 – Draw a Pattern Using print()

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```
print("*")
print("**")
print("***")
print("****")
print("*****")
```

Output

```
*
**
***
****
*****
```

Practical Activity 5 – Create a Welcome Message

```
print("**********************")
print(" Welcome to Python ")
print(" Young Technologist ")
print("**********************")
```

Classroom Activity

Predict the Output

Program 1

```
print("Python")
print("Programming")
```

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Answer

Python

Programming

Program 2

print("I Love Coding")

Answer

I Love Coding

Quick Quiz

1. Python is a:

A. Game

B. Programming Language ✅

C. Browser

D. Printer

2. Who created Python?

A. Bill Gates

B. Guido van Rossum ✅

C. Steve Jobs

D. Tim Berners-Lee

3. Which function displays output?

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A. show()

B. print() ✅

C. write()

D. input()

4. Which symbol starts a comment?

A. @

B. #

C. # ✅

D. $

5. Which program prints Hello?

A.

```
print("Hello")
```

✅ Correct

Home Assignment

1. Write a Python program to print your:
    - Name
    - School
    - Class
    - Favorite hobby
2. Create a pattern using print() with stars (*).

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3. Write five reasons why you want to learn Python.
4. Write the definition of Python in your own words.
5. Draw a flow diagram showing: **Student → Python Program → Computer → Output**.

Lesson Summary

Today we learned:

- What Python is.
- The history of Python.
- Why Python is popular.
- Real-world applications of Python.
- How to open Python IDLE or an online editor.
- How to write and run the first Python program.
- The print() function.
- Python comments.
- Basic syntax rules.
- Practical coding exercises using print().

Learning Outcomes

After completing this lesson, students will be able to:

- Define Python and explain its importance.
- Describe where Python is used in everyday life.
- Open a Python programming environment.
- Write and execute simple Python programs using print().

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- Use comments to explain code.
- Follow basic Python syntax rules.
- Build confidence to begin learning variables, input, and data types in the next lesson.

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